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MILITARY SPECIFICATION

PLATOON EARLY WARNING SYSTEM AN/TRS-2( )

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Platoon Early Warning System (PEWS) AN/TRS-2( ) which is used to provide early warning of approaching personnel and vehicles to platoon and outpost size units. (See 6.6) The PEWS consists of the following:

a. Ten (10) each Detector, Anti-Intrusion DT-577( )/TRS-2. (This is a Seismic/Magnetic type sensor.)

b. Two (2) each Monitor Receiver, Radio R-1808( )/TRS-2. (This is the receiver/display unit.)

c. Two (2) each Sensor Interface Wire Link MX-9738( )/TRS. (This unit is the Wire Adapter Module used with the receiver when field wire is employed instead of RF, for one or more of the sensors.)

d. Two (2) each Case, Platoon Early Warning System CY-7524 ( )/TRS-2. (This is the carrying bag for the PEWS.)

e. Two (2) each Stake, Grounding, for Receiver.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Material Handling, Wood Stringer Construction 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-wall
PPP-B-676	Box, Setup
PPP-C-843	Cushioning Material, Cellulosic
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Non-Metallic (and Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	Tape, Pressure Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-B-131	Barrier Materials, Watervaporproof, Flexible, Heat-Sealable
MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-52798	Enamel, Alkyd, Camouflage

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STANDARDS

FEDERAL

FED-STD-356	Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Test, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

DRAWINGS

Electronics Command

DL-SM-B-783138	Receiver, Radio R-1808( )/TRS-2
DL-SM-B-783140	Detector, Anti-Intrusion DT-577 ( )/TRS-2
DL-SM-B-783144	Sensor, Interface, Wire Link MX-9738( )/TRS-2
DL-SM-B-783146	Case, Platoon, Early Warning System CY-7524( )/TRS-2
SM-D-783288	Code Plug Assembly
SM-A-783435	Acceptance Test Procedure DT-577( )/TRS-2
SM-A-783437	Acceptance Test Procedure R-1808( )/TRS-2
SM-A-783438	Acceptance Test Procedure MX-9738( )/TRS-2

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SM-A-783439	False Activation, Test Procedure DT-577( )/TRS-2.
SM-A-783440	Test Procedure (Group B), R-1808( )/TRS-2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer. Both the title and the number should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 First Article. When specified by the contract, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The AN/TRS-2( ) shall be constructed in accordance with the requirements of this specification and the following drawings:

DL-SM-B-783140	Detector, Anti-Intrusion DT-577( )/TRS-2
DL-SM-B-783138	Receiver, Radio R-1808( )/TRS-2
DL-SM-B-783144	Sensor Interface Wire Link MX-9738( )/TRS-2
DL-SM-B-783146	Case, Platoon Early Warning System CY-7524( )/TRS-2

3.3 Parts, Materials and Processes; General. In addition to the requirements of this specification and the drawings specified in 3.2, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Finish, Protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes the finish of hardware such as handles, hinges, screws, etc., and necessary touchup after mounting. The final paint film on Type I surfaces shall be lusterless forest green in accordance with MIL-E-52798(ME). All external surfaces to be coated/treated to remove shiny surface.

3.5 Marking.

3.5.1 General. Marking shall conform to MIL-M-13231. (See 4.4). Front panel marking shall be Group I as described in that specification.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial Numbers. The following items shall have serial numbers:

Detector, Anti-Intrusion - DT-577( )/TRS-2

Receiver, Radio - R-1808( )/TRS-2

Sensor Interface Wire Link - MX-9738( )/TRS-2

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3.6 Service Conditions.

3.6.1 Specified Performance. For purposes of service conditions, the required specified performance shall be as follows:

DT-577( )/TRS-2

Requirement Paragraph	Test Paragraph	Measurement
3.7.1.1	4.7.1.1	Power Consumption Test
3.7.1.2	4.7.1.2	Radio Frequency (RF) Power Output Test
3.7.1.3	4.7.1.3	RF Carrier Frequency Stability Test
3.7.1.6	4.7.1.5	Target Detection (Bench Test)

R-1808( )/TRS-2

Requirement Paragraph	Test Paragraph	Measurement
3.7.2.1	4.7.2.1	Sensitivity Test
3.7.2.2	4.7.2.2	Local Oscillator Frequency Test
3.7.2.9	4.7.2.9	Receiver Function and Display Test

MX-9738( )/TRS-2

Requirement Paragraph	Test Paragraph	Measurement
3.7.3.1	4.7.3.1	Wire Link Circuit

3.6.2 High Temperature. (See 4.8.2)

3.6.2.1 Operating. The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +51.7°C (+125°F) plus the effects of solar radiation at a rate of 360 BTU/ft^2/hr.

3.6.2.2 Storage and Transportation. The equipment (or assembly) shall withstand exposure to ambient air temperatures as high as +71°C (160°F).

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3.6.3 Low Temperature. (See 4.8.1)

3.6.3.1 Operating. The equipment shall be operable without degradation in specified performance at ambient temperatures as low as -31°C (-25°F).

3.6.3.2 Storage and Transportation. The equipment shall withstand exposure to ambient air temperatures as low as -57°C (-70°F).

3.6.4 Altitude. (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.6.5 Immersion. (See 4.8.4) The equipment shall be immersion proof to a covering depth of 3 feet of water for a period of not less than 2 hours.

3.6.6 Dust. (See 4.8.5) The equipment (or assembly) shall withstand, in both operating and non-operating condition, exposure to fine dust particles with wind speeds of 17 knots.

3.6.7 Shock, Bench-Handling. (See 4.8.6) The equipment shall withstand shocks encountered during servicing and when tested in accordance with 4.8.6 and shall show no degradation in specified performance or suffer any physical damage.

3.6.8 Vibration. (See 4.8.7) The equipment shall withstand vibration encountered during transportation and when tested in accordance with 4.8.7 shall show no degradation in specified performance or suffer any physical damage.

3.6.9 Humidity. (See 4.8.8) The equipment (or assembly) shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.6.10 Fungus. (See 4.8.9) After the fungus test of 4.8.9, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six (6) minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.4.3) (which includes branching (see 6.4.1) and sporulation (see 6.4.5) on or within each cubic foot or fraction thereof, of equipment assembly volume.) Isolated instances of partial tubular germination (see 6.4.6) shall not be included in this evaluation.

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3.6.11 Salt Fog. (See 4.8.10) After the salt fog test of 4.8.10, the equipment, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation such as flaking, pitting, blistering, or loosening of finish or metal surface, or exfoliation (see 6.4.2) of metal. Electrical operation is not required for this test.

3.6.12 Individual Drop. (See 4.8.11) The DT-577, R-1808, and MX-9738 shall withstand shocks encountered during operation.

3.7 Electrical Performance Requirements.

3.7.1 Detector, Anti-Intrusion, DT-577( )/TRS-2.

3.7.1.1 Power Consumption. (See 4.7.1.1) The equipment standby power consumption shall not exceed 1.0 ma max. The current drain on the battery during transmission shall not exceed 200 ma.

3.7.1.2 RF Power Output. (See 4.7.1.2) The RF power level radiating from the sensor transmitter antenna shall be not less than 100 mW at 5.5 VDC input. (See 6.9)

3.7.1.3 RF Carrier Frequency Stability. (See 4.7.1.3) The frequency stability over the entire operational temperature range shall be within ±40 PPM during the transmission period.

3.7.1.4 Modulation Deviation. (See 4.7.1.8) The deviation around the carrier frequency shall be essentially symmetrical and shall not be less than ±8 kHz nor greater than ±14 kHz for both F_1 and F_2.

3.7.1.5 Tone Generator and Clock Frequency. (See 4.7.1.4) The RF message shall contain all the information necessary for the receiver to display the sensor identification (ID) number and the target's classification (personnel or vehicle). The modulation techniques used shall be Pulse Width Modulation/FSK. The RF carrier shall be frequency modulated by two (2) audio frequency tones to perform the frequency shift function. The low frequency tone, called F_1, shall be 1500 ±30 Hz, and the high frequency tone, called F_2, shall be 1800 ±45 Hz. The RF message shall consist of a 10-bit word. The bit period shall be divided into three (3) equal segments. The first segment shall always be modulated by F_2. The third segment shall always be modulated by F_1. The second segment shall be modulated by F_1 if the bit is a "0", or F_2 if the bit is a "1". The message shall be composed of 1's and 0's organized in the manner described. The 10-bit message shall be preceded by a 30 ±10 ms period, wherein the RF carrier is frequency modulated by F_1. This 30 ms period permits the

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receiver time to settle. Each 10-bit message shall be assembled as indicated below in the order shown:

a. One (1) bit for marker

b. Three (3) bits for PEWS area

c. Four (4) bits for the sensor ID number

d. One (1) bit for odd "1's" parity

e. One (1) bit for target classification

The bit rate shall not be less than 30 bits per second nor greater than 37 bits per second. Each sensor in a particular PEWS set shall have the same address but a unique ID number (1-16). The adjustment of each sensor to its address and ID number shall be accomplished by a Code Plug Assembly, SM-D-783288. This assembly shall be installed at the factory uncoded, and shall be capable of being replaced with another code plug.

3.7.1.6 Target Detection (See 4.7.1.5 and 6.4.8).

3.7.1.6.1 Personnel Detection. (See 4.7.1.5) The sensor shall detect up to three (3) persons walking at a distance of 10 meters from the emplaced sensor with a 75% or greater probability of detection for a single person and an 85% or greater probability of detection for two (2) or three (3) persons. (See 6.10)

3.7.1.6.2 Vehicle Detection. (See 4.7.1.5) The sensor shall be required to detect a vehicle as small as a 5/4 ton truck traveling at a speed of 10 mph. The equipment shall detect any tracked vehicle traveling at speeds greater than 5 mph. The probability for detection for wheeled vehicles or tracked vehicles shall be at least 85% at a distance of 10 meters from the emplaced sensor. (See 6.11)

3.7.1.7 Target Classification. (See 4.7.1.6) The sensor shall discriminate correctly between persons and vehicles 85% of the time when a detection is made and when the person or vehicle is within 10 meters of the sensor. The classification output shall be in accordance with Table I.

3.7.1.8 RF Carrier Frequency. (See 4.7.1.3) The sensor carrier frequency shall be within $\pm$25 PPM at ambient temperatures.

3.7.1.9 False Activation. (See 4.7.1.9) A sensor shall not falsely activate at a rate of three (3) or more times per hour for any one hour period during any 72 hour period of continuous operation and no more than 10 total activations during the 72 hour period. (See 6.4.7)

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3.7.1.10 Test Switch. (See 4.7.1.10) The sensor shall be provided with a test switch to activate the encoder-transmitter circuits in order to test the sensor's RF transmission.

3.7.2 Receiver Radio, R-1808( )/TRS-2. (See 4.7.2)

3.7.2.1 Receiver Sensitivity. (See 4.7.2.1) The receiver shall be sensitive enough to operate with an RF input of .4 μV. The receiver bandwidth shall not be greater than 40 kHz.

3.7.2.2 Local Oscillator Frequency. (See 4.7.2.2) The receiver's crystal controlled local oscillators shall have an accuracy of ±30 PPM of the designated frequency.

3.7.2.3 Spurious Response. (See 4.7.2.3) The receiver shall have no spurious responses to input signals which are less than 35 dB above 1.0 μV at the receiver input terminals. Also, there shall be no more than one (1) spurious response to input signals which are less than 40 dB above 1.0 μV at the receiver input terminals.

3.7.2.4 Desensitization. (See 4.7.2.4) When receiving a 1.0 μV signal, the receiver shall function properly in the presence of an undesired signal (except those frequencies resulting in a spurious response having an EMF of 20 mV and and being ±10 percent or more removed in frequency from the receiver operating (tuned) frequency.

TABLE I
TRUTH TABLE FOR SENSOR

INPUTS			
Seismic		Magnetic	Classifier
FSP	Veh		Output
0	0	X	None
0	X	0	None
0	1	1	C
1	0	0	P
1	0	1	P

NOTE: X means that the detector output can be either a "1" or a "0".
1 means that the detector has been activated.
0 means that the detector has not been activated.
C means that a vehicle classifier is transmitted.
P means that a personnel classifier is transmitted.
"None" means that no message is transmitted.

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3.7.2.5 IF Rejection. (See 4.7.2.5) The IF rejection ratios shall be at least 50 dB.

3.7.2.6 Image Rejection. (See 4.7.2.6) The image rejection ratios of the receiver shall be greater than 45 dB.

3.7.2.7 Local Oscillator Radiation. (See 4.7.2.7) The radiation from the local oscillators within the receiver shall produce an output no greater than 100 µV at the antenna terminals when the receiver is operating at any frequency within its tuning range and as measured across a 50 ohm load.

3.7.2.8 Power Consumption. (See 4.7.2.8) The receiver current drain shall not exceed 6 mA, exclusive of lighting the display, and 25 mA total at 9 VDC input. (See 6.9).

3.7.2.9 Receiver Function and Display. (See 4.7.2.9) The receiver shall perform all functions necessary to display the correct areas, ID's and classifications and shall provide a visual or audible alarm or both. A dot shall indicate the latest activation followed by a **sequential** display of previous activations until terminated by a reset button. A manual control shall be provided with the audible alarm so that the alarm tone can be adjusted to zero output. The receiver shall monitor and indicate a low battery voltage of less than 5.5 volts. On any one frequency, the area selection switch shall provide a means for selecting up to eight (8) discrete areas of operation.

3.7.3 Wire Adapter Module MX-9738( )/TRS-2.

3.7.3.1 Wire Link Circuit (see 4.7.3.1). The wire adapter module shall provide a test position to determine if the wire link is operable. An open or shorted line under test shall be indicated by a blinking light. A steady light shall indicate a good line when in the test position.

3.7.3.2 Power Consumption (see 4.7.3.2). The wire module will share the same power supply used by the receiver. The power drain of the wire module shall be less than 1 mA except when in the test mode.

3.7.4 Wire Communication (see 4.7.4). The sensors shall be capable of transmitting activations to the receiver through a wire link at a distance of 1500 meters. The wire link shall use WO-36 wire or a lightweight wire with no more than 16 dB total attenuation at 2 kHz. The wire link is defined as one sensor connected to the receiver over a single wire pair.

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3.8 Burn-in. Each PEWS shall be capable of meeting the requirements herein, without subsequent processing after subjection to the burn-in procedure of 4.6.

3.9 Electromagnetic Interference (see 4.9). The equipment shall meet the emission and susceptibility requirements of Table VI, Ground C-E Equipment and Component Items of Equipment, of MIL-STD-461A, Notice 4, as defined herein for nonsheltered equipments:

(1)	(2)	RE02		(1)	(3)	RS03.2
(1)	(2)	RE02.1				
	(4)	CE06				

(1) Tests shall be performed using wire and RF communication links.

(2) Radiated switching transients shall comply with the limits of para. 6.13.4 of MIL-STD-461A, Notice 4.

(3) The equipment shall be immersed in an electric field of 5 volts/meter over the 45 kHz to 1.53 GHz frequency range. The following modulation characteristics shall be used on the interfering signal:

45 kHz to 30 MHz	1 kHz, 30% AM sine wave
30 MHz to 200 MHz	1.8 kHz FM ±10 kHz deviation
200 MHz to 1.53 GHz	1.8 kHz, 100% AM square wave

(4) The RF power levels of spurious and harmonic emissions above the third harmonic conducted from the antenna, relative to the power of the fundamental frequency shall be at -55 dB for the sensor at the nominal RF power level of 200 milliwatts. For frequencies equal to or below the 3rd harmonic the limit of -33 dB shall apply.

3.10 Interchangeability (see 4.11). The equipment shall be designed such that parts, components, and assemblies will be functionally, electrically, and mechanically interchangeable at all maintenance levels without resorting to selective assembly. Interchangeability for electronic equipment shall be in accordance with MIL-STD-454, Requirement 7.

3.11 System Weight (see 4.12). The overall weight of the system without batteries and grounding stake shall not exceed 9.6 kg. (21.0 lb)

3.12 Workmanship (see 4.13). The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

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3.13 Reliability (see 4.10). The system, as defined by Note 1.c of Table II, shall have a (θ_o) specified Mean-Time-Between-Failure (MTBF) of 350 hours while operating under the following conditions:

a. Temperature - 43°C (110°F).

b. Duty cycle - Sensors - Continuous operation in a standby state at 4 activations per hour.

- Receiver - Continuous operation in a standby state at 40 activations per hour.

3.14 Systems Safety Engineering.

3.14.1 Personnel Hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454 and MIL-STD-1472, "Labeling" (5.5) and "Hazards and Safety" (5.13).

3.14.2 Edge Rounding Exposed. Exposed edges shall be rounded to a minimum radius of .004 inch (.1mm) and exposed corners to a minimum of 0.1 inch (2.5mm).

3.14.3 Radioactive Materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.14.4 Chemical Safety. The system design shall insure that no materials or components are used which during any phase of the life cycle will evolve toxic gases, vapors, fumes or droplets. Materials which support combustion or are capable of causing an explosion shall not be used. Dissimilar metal combinations shall be in accordance with requirement 16 of MIL-STD-454 for Class II environment.

3.15 Human Factors Engineering. Design configuration as affecting operation, maintainability and handling shall be maintained through compliance of configuration criteria to the requirements of MIL-STD-1472.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3): Does not include preparation for delivery.

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections:

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.14).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 The contractor shall furnish the number of first article units in accordance with the following:

a. Eighteen (18) each R-1808( )/TRS-2.

b. Seventy-four (74) each DT-577( )/TRS-2.

c. Eighteen (18) each MX-9738( )/TRS-2.

d. Five (5) each CY-7524( )/TRS-2.

4.3.2 First Article Inspection. The first article inspection shall consist of inspections in Table II, and shall be performed in the order specified therein.

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4.3.3 First Article Data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspections Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Parts, materials, and processes; general	3.3
Finish, protective	3.4
Marking	3.5

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. The contractor shall perform the inspections specified in 4.4, and 4.5.1 through 4.5.5. Each unit on contract shall be subjected to the Burn-in of 4.6 prior to acceptance inspection. This does not relieve the contractor of his responsibility for performing any additional inspection which is deemed necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each DT-577( )/TRS-2, R-1808( )/TRS-2, and MX-9738( )/TRS-2 shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. A lot shall consist of units of one type of individual unit, such as the R-1808. Factors of lot composition not defined herein, or in the contract or in the purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table III.

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TABLE II

FIRST ARTICLE INSPECTION

Inspections	Reqt. Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Units 6-10
1. Inspections covered by subsidiary documents.	3.3, 3.4, 3.5	4.4	Inspections to be performed on all units.					
2. Burn-In	3.8		"	"	"	"	"	"
3. Group A Inspection	(See Table III)		"	"	"	"	"	"
4. Group B Inspection	(See Table IV)		"	"	"	"	"	"
5. Group C Inspection								
(Environmental)								
Subgroup 1	(See Table V)		X					
Subgroup 2	(See Table V)			X				
Subgroup 3	(See Table V)				X			
(Electrical)								
Subgroup 4	(See Table V)					X		
Subgroup 5	(See Table V)						X	
6. Group D Reliability	3.13							X

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TABLE II

FIRST ARTICLE INSPECTION (Cont'd)

NOTES:

1. a. For first article units 1 through 4, one unit is defined as follows:

   One (1) each DT-577( )/TRS-2
   One (1) each R-1808( )/TRS-2
   One (1) each MX-9738( )/TRS-2

   First article unit 3 shall also include the CY-7524( )/TRS-2

   b. For first article unit 5, one unit is defined as follows:

   Twenty (20) each DT-577( )/TRS-2
   Four (4) each R-1808( )/TRS02
   Four (4) each MX-9738( )/TRS-2
   Four (4) each CY-7524( )/TRS-2

   c. For first article units 6 through 10, one unit is defined as follows:

   Ten (10) each DT-577( )/TRS-2
   Two (2) each R-1808( )/TRS-2
   Two (2) each MX-9738( )/TRS-2

2. The inspections 1 through 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

3. The implementation of the inspections covered by 5 and 6 shall be expedited to satisfy contract delivery requirements.

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4.5.1.1 Order of Inspection Within Group A. Group A Inspection shall be performed in an order satisfactory to the Government.

TABLE III

GROUP A INSPECTION

Inspection	Reqt. Para.	Test Para.	AQL Major	AQL Minor
Visual and Mechanical	3.12	4.13	1.0%	4.0%
Electrical				
DT-577( )/TRS-2				
Power Consumption	3.7.1.1	4.7.1.1	1% For The Group	
RF Power Output	3.7.1.2	4.7.1.2		
RF Carrier Frequency	3.7.1.8	4.7.1.7		
Tone Generator & Clock Frequency	3.7.1.5	4.7.1.4		
Target Detection (Bench Test)	3.7.1.6	4.7.1.5		
Target Classification (Bench Test)	3.7.1.7	4.7.1.6		
*All electrical defects are major.				
R-1808( )/TRS-2				
Receiver Sensitivity	3.7.2.1	4.7.2.1	1% For The Group	
Local Oscillator Frequency	3.7.2.2	4.7.2.2		
Power Consumption	3.7.2.8	4.7.2.8		
Receiver Function and Display	3.7.2.9	4.7.2.9		
MX-9738( )/TRS-2				
Wire Link Circuit	3.7.3.1	4.7.3.1	1% For The Group	
Power Consumption	3.7.3.2	4.7.3.2		

4.5.2 Group B Inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and meet Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table I of MIL-STD-105.

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4.5.2.1 Group B Sampling Plans. The Group B sampling plans, for the AQL's listed in Table IV, shall be as follows:

AQL	Inspection Level
4%	S-4
6.5%	S-3

4.5.2.2 Order of Inspection Within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE IV

GROUP B INSPECTION

Inspection	Reqt. Para.	Test Para.	AQL
DT-577( )/TRS-2			
False Activation	3.7.1.9	4.7.1.9	4%
R-1808( )/TRS-2			
Spurious Response	3.7.2.3	4.7.2.3	4% For The Group
Densensitization	3.7.2.4	4.7.2.4	
IF Rejection	3.7.2.5	4.7.2.5	
Image Rejection	3.7.2.6	4.7.2.6	
Local Oscillator Radiation	3.7.2.7	4.7.2.7	
System Weight	3.11	4.12	6.5%
Interchangeability	3.10	4.11	6.5%

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

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4.5.3.1 Sampling for Group C Inspection. One (1) sample unit shall be selected at random for each subgroup of inspections in Table V from each 3 months production or 1000 systems whichever occurs first. Units tested in Subgroups 1 through 4 are as defined in note 1a of Table II. The unit tested in Subgroup 5 is as defined in note 1b of Table II. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed in the order shown in each subgroup of Table V.

TABLE V

GROUP C INSPECTION

Inspection	Reqt. Para.	Test Para.
Subgroup 1		
Low Temperature	3.6.3	4.8.1
High Temperature	3.6.2	4.8.2
Altitude	3.6.4	4.8.3
Leakage (Immersion) #1	3.6.5	4.8.4
Shock, Bench Handling	3.6.7	4.8.6
Individual Drop	3.6.12	4.8.11
Vibration	3.6.8	4.8.7
Leakage (Immersion) #2	3.6.5	4.8.4
*Salt Fog	3.6.11	4.8.10
Subgroup 2		
Humidity	3.6.9	4.8.8
Subgroup 3		
Dust	3.6.6	4.8.5
*Fungus	3.6.10	4.8.9

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TABLE V

GROUP C INSPECTION (Cont'd)

Inspection	Reqt. Para.	Test Para.
Subgroup 4		
Electromagnetic Interference	3.9	4.9
Subgroup 5		
Bench Tests		
RF Stability	3.7.1.3	4.7.1.3
Modulation Deviation	3.7.1.4	4.7.1.8
Wire Communication	3.7.4	4.7.4
Field Tests		
Target Detection	3.7.1.6	4.7.1.5
Target Classification	3.7.1.7	4.7.1.6

* The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Conforming Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspections may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Group D Inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

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4.5.5.1 Sampling for Group D Inspection. Five (5) units as defined in Note 1c of Table II, shall be selected at random in accordance with 4.10.

TABLE VI

GROUP D INSPECTION

Inspection	Reqt. Para.	Inspection Para.
Reliability	3.13	4.10

4.5.5.2 Group D Failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.6 Reinspection of Conforming Group D Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Burn-In. Every item (DT-577, R-1808, MX-9738) before undergoing Group A testing shall be subjected to a 24 hour (on-time) burn-in under the following conditions:

a. Vibration. Each item shall be mounted on a vibration table and vibrated (non-operating) for 20 minutes at 2.2 G ±10% acceleration at any nonresonant frequency between 20 and 60 Hz.

b. Operational period. Subsequent to vibration, each item shall be operated for 24 hours under the conditions of 4.10.1. The last 12 hours of this operating period shall be failure free.

c. As a minimum, each item shall be checked for satisfactory operation at the beginning and end of the 12 hour failure free period.

d. The final 12 hours of operation shall be failure free. The failure definition of 4.10.2 shall apply. If a failure occurs during the failure free period, the equipment shall be repaired and returned to burn-in at the beginning of the 12 hour failure free period. Burn-in shall continue until a 12 hour failure free period has been achieved.

4.7 Electrical Performance Tests.

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4.7.1 Detector, Anit-Intrusion, DT-577( )/TRS-2.

4.7.1.1 Power Consumption. Sensor power consumption shall be measured with 9.0 VDC input in accordance with SM-A-783435.

4.7.1.2 RF Power Output. The RF power level radiating from the sensor transmitter antenna shall be measured with 5.5 and 9.0 VDC inputs in accordance with SM-A-783435.

4.7.1.3 RF Carrier Frequency Stability. The sensor transmitter frequency and modulation tones shall be measured at -31°C (-25°F), 71°C (160°F), and room ambient temperatures in accordance with SM-A-783435.

4.7.1.4 Tone Generator and Clock Frequency. The accuracy of the clock frequency and modulation tones shall be measured on the Data Link Board prior to sensor assembly, in accordance with SM-A-783435.

4.7.1.5 Target Detection. Target detection tests shall be conducted as electrical laboratory test (Group A) in accordance with SM-A-783435, and as an operational field test (Group C) in accordance with a government approved test plan.

4.7.1.6 Target Classification. A target classification test shall be conducted as an electrical laboratory test (Group A) in accordance with SM-A-783435, and as an operational field test (Group C) in accordance with a government approved test plan.

4.7.1.7 RF Carrier Frequency. The sensor shall be tested for modulation deviation in accordance with SM-A-783435.

4.7.1.8 Modulation Deviation. The sensor shall be tested for modulation deviation in accordance with SM-A-783435.

4.7.1.9 False Activations. The sensor shall be tested for false activations in accordance with SM-A-783439.

4.7.1.10 Test Switch. The sensor shall be tested for proper test switch operation in accordance with SM-A-783435.

4.7.2 Receiver, Radio R-1808( )/TRS-2.

4.7.2.1 Receiver Sensitivity. The receiver sensitivity shall be measured in accordance with SM-A-783437.

4.7.2.2 Local Oscillator Frequency. The receiver's local oscillator frequency shall be measured in accordance with SM-A-783437.

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4.7.2.3 Spurious Response. The receiver spurious response shall be measured in accordance with SM-A-783440.

4.7.2.4 Desensitization. The receiver's loss of sensitivity in the presence of strong signals on nearby frequencies shall be measured in accordance with SM-A-783440.

4.7.2.5 IF Rejection. The receiver IF rejection ratio shall be measured in accordance wtih SM-A-783440.

4.7.2.6 Image Rejection. The receiver image rejection ratio shall be measured in accordance with SM-A-783440.

4.7.2.7 Local Oscillator Radiation. The receiver local oscillator radiation shall be measured at the antenna terminals in accordance with SM-A-783440.

4.7.2.8 Power Consumption. The receiver power consumption shall be measured with 9.0 VDC input in accordance with SM-A-783437.

4.7.2.9 Receiver Function and Display. The receiver shall be tested for visual and audible alarms in accordance with SM-A-783437.

4.7.3 Sensor Interface Wire Link, MX-9738( )/TRS-2.

4.7.3.1 Wire Link Circuit. The wire link shall be tested for proper operation in accordance with SM-A-783438.

4.7.3.2 Power Consumption. The wire link power consumption shall be measured with 9.0 VDC input in accordance with SM-A-783438.

4.7.4 Wire Communication. The system shall be tested with WD-36 wire to meet the requirement in accordance with SM-A-783438.

4.8 Service Condition Tests.

4.8.1 Low Temperature. The equipment shall be tested in accordance with Method 502.1, Procedure I of MIL-STD-810. The storage temperature shall be -57°C (-70°F) and shall be maintained for a period of not less than two (2) hours following stabilization of the test item. Low operating temperature shall be -31°C (-25°F).

4.8.2 High Temperature. The equipment shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. The highest operating temperature shall be 71°C (160°F). This includes the effects of solar radiation.

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4.8.3 Altitude. The equipment shall be subjected to the test of Method 500.1, Procedure I of MIL-STD-810 except that the step 2 operating altitude shall be 10,000 feet. The equipment shall withstand 40,000 feet (non-operating) exposure for not less than 1 hour duration prior to the operational portion of step 2. Altitude rate changes should not exceed 2,000 feet per minute.

4.8.4 Immersion. The individual units shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810. Following the immersion period, there shall be no evidence of leakage and the equipment shall be operated to insure that no degradation in performance has resulted. This test shall be performed prior to shock and following vibration, shock and individual unit drop of the same test item.

4.8.5 Dust. The individual units shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810.

4.8.6 Shock Test, Bench Handling. The individual units shall be subjected to the tests of Method 516.2, Procedure V of MIL-STD-810.

4.8.7 Vibration. The individual units shall be subjected to Method 514.2, Procedure XI, Part 2 of MIL-STD-810.

4.8.8 Humidity. The individual units shall be subjected to the test of Method 507.1 of MIL-STD-810, Procedure III. In step 6, test measurements shall be taken during the last five (5) hours of the last cycle. In step 7 the battery compartment only shall be opened.

4.8.9 Fungus. The individual units and the carrying bag shall be subjected to the fungus test specified in Method 508.1, Procedure I, of MIL-STD-810. There shall be abundant growth colonization (see 6.4) on 50 percent or more of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.6.10 shall constitute failure of this test.

4.8.10 Salt Fog. The individual units shall be subjected to the salt fog test specified in Method 509.1, Procedure I, of MIL-STD-810. No corrosion products shall be deposited in the test item by the facility. Inability of the equipment to meet the requirements of 3.6.11 shall constiutue failure of this test.

4.8.11 Individual Drop. The individual units shall be subjected to the test of Method 516.2, Procedure II, of MIL-STD-810. Units shall be tested without carrying cases.

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4.9 Electromagnetic Interference (EMI). The units as defined in 4.5.3.1 shall be tested to meet the EMI requirements of 3.9. Test setups and procedures shall be in accordance with the requirements of MIL-STD-462, Notice 3 as implemented by a contractor prepared, Government approved, test plan.

4.10 Reliability Testing. The contractor shall conduct a Reliability Qualification Test as part of First Article Testing. This test shall be in accordance with Test Plan XX of MIL-STD-781 and a contractor prepared, Government approved Reliability Test, Demonstration and Evaluation Plan. The contractor shall also conduct periodic Reliability Acceptance Tests, on samples taken from each 1000 systems produced, according to Test Plan XXI of MIL-STD-781 and the approved test plan. The specified MTBF for all reliability tests shall be 350 hours. A sample size of five (5) systems shall be used for each test. No single system "on" time shall be less than one-half the average operating time of all systems on test. Three (3) systems shall be operated in the RF mode and two (2) systems shall be operated in the wire mode throughout the reliability test.

4.10.1 Reliability Test Condition. The equipment shall be tested under the following conditions:

a. Temperature - 43°C (110°F)

b. Duty Cycle - Continuous with four (4) activations per hour per sensor (40 monitor-receiver activations per hour).

4.10.2 Failure Definition. A failure is defined as any inability of the equipment to perform its required function. This includes any event which causes departure from performance as required in this specification and in the Government approved test plan. During the test, proper instrumentation shall be provided to ensure recognition of equipment failures.

4.11 Interchangeability. The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

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4.12 Weight. The system shall be weighed to determine compliance with 3.11.

4.13 Visual and Mechanical Inspection. The equipment shall be examined to insure freedom from the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with portions of 3.14 (System Safety Engineering) which can be determined visually.

4.14 Quality Conformance Inspection of Preparation for Delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.2C).

5.1.1 Level A.

5.1.1.1 Cleaning. Platoon Early Warning System AN/TRS-2( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Platoon Early Warning System AN/TRS-2( ) shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative Application. No preservative required.

5.1.1.4 Unit Packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical Literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Accessory Items. Each accessory item shall be individually unit packed method III as follows: wrap each item with paperboard conforming to PPP-P-291, type 2. Secure each wrap with tape conforming to PPP-T-45, type III. Place the wrapped items bearing the same stock number, in the quantities as specified within a close-fitting box conforming to PPP-B-566 or PPP-B-676, variety 2. Close the box in accordance with appendix of the box specification.

5.1.1.4.3 Major Components. Each major component shall be individually unit packed method IIc. Wrap each item with barrier material conforming to MIL-B-121, type II. Cushion each wrapped item with barrier material conforming to PPP-C-843, type II, as required to protect the item. Place desiccant, conforming to MIL-D-3464 in the amount specified in MIL-P-116, on top of the wrapped item. Place the cushioned item within a close-fitting barrier bag fabricated of material conforming to MIL-B-131, class 2, with sufficient allowance in one direction to permit two subsequent resealings. Heat seal the closure, leaving a small opening for exhausting air. Exhaust the entrapped air and complete the seal.

5.1.1.4.4 Platoon Early Warning System AN/TRS-2( ). Accessory items and major components, unit packed as specified in 5.1.1.4.2 and 5.1.1.4.3 shall be consolidated within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with appendix of the box specification.

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5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1, except the method IA-8 shall be substituted for method IIc.

5.1.3 Commercial Preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Platoon Early Warning System AN/TRS-2( ), unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. A quantity of Platoon Early Warning System AN/TRS-2( ), unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial Packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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5.3.2 Commercial Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

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6. NOTES

6.1 System Description. The PEWS is a lightweight, self-powered, portable intrusion detection device designed to be used by small military units or groups such as patrols, platoons or squads. The sensors have been designed for hand emplacement and unattended operation in forward combat zones. The system incorporates self-contained sensors capable of detecting personnel and vehicles at ranges up to 10 meters from the emplanted sensors and is able to distinguish between personnel and vehicle intrusions. The sensor consists of a combination of detectors. The total number of sensors in a PEWS is ten (10). The type of sensor comprising the PEWS configuration is a Seismic/Magnetic Combination. The detector combination required to assure a high probability of detection of walking men and vehicles and to classify between them. Upon the detection of an intrusion, the sensor, after classifying the intruder (as to man or vehicle), communicates the detection event and the classification data to a remote monitor receiver/display by means of either an RF link or a wire link. The communication option is selectable by means of a switch on the PEWS sensor. The communications message also contains the ID number of the activated sensor. The remote monitor receiver/display (to be called the "receiver" hereafter) is capable of receiving the RF transmission directly and displaying, by means of lights, the ID number of the activated sensor and the classification of the intruder. The receiver shall perform the same display functions when a wire link is used; however, a wire adapter module will be used to interface the communication wire to the receiver. The receiver is completely self-contained when used in the RF mode. The wire module accommodates nine (9) pairs of wires although five (5) pairs would normally be used for each receiver. The wire module contains all the circuits, controls, and indicator lamps necessary to monitor the condition of the wires used to connect the sensor to the wire module. The wire module shall use the same power supply used by the receiver when connected to it. The receiver is protected against high voltages when used with the wire module by use of a ground stake. Operator protection is also provided by means of the grounding stake. The grounding stake is to be used only when the wire link is used. The receiver is capable of operating in the RF mode and the wire link mode simultaneously.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. PEWS frequency channels. The RF communications link is capable of operating in any frequency channels within the 138 to 153 MHz band. This channel is set by the manufacturer and shall be specified in the contract.

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c. Level A, Level B or Commercial preservation and packing (see Section 5).

d. The specific paragraphs of Section 5 which are applicable to the particular procurement.

e. First article. See paragraph 4.3.1 for number of first article unit/systems required.

f. Marking and shipping of samples.

g. Place of final inspection.

h. When the rough handling test is required.

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Definitions.

6.4.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4.3 Growth Colonization. Growth colonization is a mass of individual plants, generally of one (1) species, living together; or a group of hyphae which is formed from one (1) spore or cell and may be one (1) individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

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6.4.4 Microbial Growth. Microbial growth is the growth of very minute organisms, which when present in large amounts, may provide a colony visible to the naked eye.

6.4.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.4.6 Tubular Germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.4.7 False Activation. A false activation is an activation of the sensor not caused by any external stimulus which produces an event on the receiver. An event on the receiver is defined as a visual or aural indication due to a signal from the sensor. External stimulus is defined for the purposes of this specification to include bonafide targets such as persons and vehicles, animals of any size, insects, rain, hail, snow, sleet and wind in excess of 10 MPH. Changes in temperature and humidity and winds of 10 MPH or less are not considered external stimuli.

6.4.8 Detection. Detection is a sensor indication of a valid target whether classification is correct or not.

6.5 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3)

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/TRS-2W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.7 Verification Inspection. Verification by the Government will be the amount deemed necessary to determine compliance with the contract.

6.8 Size of the PEWS. (Exclusive of antennas and sensor stakes.)

Component	Length(cm)	Width(cm)	Height(cm)	Volume(cucm)
DT-577	16.8 (6.6")	9.7 (3.8")	6.4 (2.5")	1050 (64 cu in)
Receiver	19.8 (7.8")	11.9 (4.7")	4.8 (1.9")	1130 (68.9 cu in)
Wire Module	5.1 (2")	11.9 (4.7")	6.9 (2.7")	420 (25.6 cu in)
Carrying Bag/System	To be specified at a later date.			

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6.9 Equipment Power Source. The equipment uses the BA-90 or BA-3090 battery. DT-577 uses one battery and the R-1808 uses two (2) batteries.

6.10 Detection of Four (4) or More Persons. The sensor is expected to detect four (4) or more persons, but a requirement for probability detection is not specified, since this was not a user requirement.

6.11 Detection of Vehicles. The sensor is expected to detect vehicles of any size, but since this is not a user requirement, no probability of detection is specified for vehicles smaller than 5/4 ton.

Custodian:
Army - EL

Preparing Activity:
Army - EL
Project # 5895-A259

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-P-49115A(EL) PLATOON EARLY WARNING SYSTEM AN/TRS-2( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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